UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Spring 2024

Roll No. ....

Paper: Electronics-II **Course Code: PHYS-3702** **Time: 3 Hrs. Marks: 60**

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. **(15x2=30)**

i. Convert $(001010100100)_2$ to hexadecimal.
ii. Write 279 in excess 3 code. 010110101100
iii. Is NAND gate associative or not?
iv. Differentiate between amplifier and oscillator.
v. Convert (10011100)gray code to binary. 11101000
vi. Convert $(354)_8$ to hexadecimal.
vii. Why does the Class C amplifier prefer over other amplifiers?
viii. Convert the given expression into standard form.

$$X = (A+C)(A+B)$$

ix. Differentiate between multivibrator and flip-flop.
x. Write down the expression for power gain of common collector amplifier in terms of ratio of resistances.
xi. What is a bandwidth of an amplifier when upper critical frequency=5kHz and lower critical frequency =250Hz?
xii. List the amplifier stages in a typical operational amplifier.
xiii. What are the applications of operational amplifier?
xiv. At what point is a class C amplifier normally biased?
xv. What is push-pull operation in class B amplifier?

Q.2. Answer the following questions. **(3x10=30)**

I. Analyze the bipolar junction transistor amplifier for total low frequency response.
a. Determine the lower critical frequency and phase shift of input RC circuit.
b. Determine the lower critical frequency and phase shift of output RC circuit.
c. Determine the lower critical frequency of bypass RC circuit.
d. Describe a Bode plot.

II. Briefly discuss the ways to connect negative feedback with operational amplifier. What are the benefits of negative feedback in operational amplifiers?

III. What is oscillator? Discuss working principle of oscillator. Name the two types of oscillators.

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Electronics II
Past paper (Spring 2023 + 2024)
Short Questions

Question no: 01

Convert $(001010100100)_2$ to hexademical.

Answer:

Binary to hexadecimal:

→ Group the binary digits into 4-bit chunks from right to left:

0010 1010 0100

Convert each group to hexadecimal equivalent

0010 = 2 , 1010 = A , 0100 = 4

So, by combining hexadecimal digits;

$(001010100100)_2 = 2A4_{16}$

Question no: 02

Write 279 in excess 3 code.

Answer:

Convert each digit 279 to 4-bit binary

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2 = 0010 , 7 = 0111 , 9 = 1001

Now Add 3(0011) to each digit

Excess - 3

0010 + 0011 = 0101

0111 + 0011 = 1010

1001 + 0011 = 1100

So, 279 in excess - 3 code is

0101 1010 1100

Question no: 03

Is NAND gate associative or not?

Aswer:

Yes the NAND gate is associative. Mathematically, this can be expressed as:

(A NAND B) NAND C = A NAND (B NAND)C)

This property allows you to rearrange the inputs and still obtain the same output, making the NAND gate a fundamental building blocks in logic circuit.

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Question no: 04

Differentiate between amplifier and oscillator.

Answer:

Amplifer	Oscillator
→ Boosts the strength of an input signal without changing its frequency.	Generate a repetitive signal without needing an input signal.
→ The output of an amplifier is a stronger version of input signal.	The output of an oscillator is a continuous periodic waveform.
→ Amplifier uses negative feedback to stabilize gain.	Oscillators use positive feedback to maintain oscillations.
Uses	**Uses**
• Audio system	• Signal generators
• Radio receivers	• RF transmitters

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Question no: 05

Convert (10011100) gray code to binary.

Answer.

Gray to Binary

Write 1st value same and then add answer value in next.

If sum is 2 then we know that 2 is equal to 10 but we write only zero as an answer.

1	0	0	1	1	1	0	0
↓	↓	↓	↓	↓	↓	↓	↓
1	1	1	0	1	0	0	0

So;

(10011100) gray code is equal to (11101000) binary digit.

Question no: 06

Convert $(354)_8$ to hexadecimal.

Answer:

Octal to hexadecimal

To convert octal number

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to hexadecimal : first convert octal into decimal.

$$(354)_8 = 3\times8^2 + 5\times8^1 + 4\times8^0$$
$$= 3\times64 + 5\times8 + 4\times1$$
$$= 192 + 40 + 4$$
$$= (236)_{10}$$

Now convert decimal number 236 into hexadecimal.

$$\begin{array}{r|l} 16 & 236 \\ 16 & 14 - 12 \\ & 0 - 14 \end{array}$$

$14 = E$ & $12 = C$

So;

$(354)_8$ in hexadecimal is EC_{16}

Question no: 07

Why does the class C amplifier prefer over other amplifiers?

Answer:

Class C amplifiers prefered over other amplifiers due to its:

High Efficiency:

Above 80% reducing heat

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dissipation

High Power Density:

Handling high power levels.

These advantages make class C amplifier suitable for applications like RF amplifier and switch-mode power supplies.

Question no: 08

Convert the given expression into standard form.

$$X = (A+C)(A+B)$$

Answer:

Apply distributive law

$$X = A(A+B) + C(A+B)$$

$$X = AA + AB + CA + CB$$

As $\because AA = A$ (idempotent law)

So;

$$X = A + AB + CA + CB$$

$\because A + AB = A$ (absorption law)

Then;

$$X = A + CA + CB$$

$\because A + CA = A$ (absorption law)

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We get;

$$X = A + CB$$

Question no: 09

Differentiate between multivibrator and flip-flop.

Answer:

Multivibrator	Flip-Flop
Multivibrator generate wave forms.	Flip-flop store data.
Multivibrators have multiple stable states.	Flip-Flop have 2 stable state (0 & 1).
Multivibrator are used in timming and pulse generation.	Flip-Flop are used in data storage and sequential logic.

Question no: 10

Write down the expression for power gain of common collector amplifier in terms of ratio of resistance.

Answer:

The power gain of common collector amplifier is given by:

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$$A_p = \frac{P_{out}}{P_{in}}$$

Since power is proportional to the square of voltage and inversely proportional to resistance.

$$A_p = \left(\frac{V_{out}}{V_{in}}\right)^2 \cdot \frac{R_L}{R_i}$$

For a common collector amplifier

Voltage gain $A_v \approx 1$

So;

$$A_p = \frac{R_L}{R_i}$$

→ R_L is load resistance

→ R_{in} is input resistance

Question no: 11

What is a band width of an amplifier when upper critical frequency = 5kHz and lower critical frequency = 250Hz

Answer:

$$f_{upper} = 5kHz = 5000Hz$$

$$f_{lower} = 250Hz$$

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$$\text{Band width} = f_{upper} - f_{lower}$$
$$= 5000 - 250$$
$$= 4750\ Hz$$
$$\text{Band width} = 4.75\ kHz$$

Question no: 12

List the amplifier stages in a typically operational amplifier?

Answer:

Typical operational amplifier consists of three stages:

Differential Amplifier Stage:

This stage amplifies the difference between two input signals.

Voltage Amplifier stage:

This stage provide additional voltage gain to the input signal

Output Stage:

This stage provides a low-impedence output and drives the load.

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Question no: 13

What are the applications of operational amplifier?

Answer:

Applications of operational amplifiers are:

- → Amplifiers (Voltage, current & power amplification)
- → Filters (Low pass, high pass, band pass)
- → Oscillators (Generate wave forms)
- → Comparators (Compare two voltages and output signal)

Question no: 14

At what point a class C amplifier normally biased?

Answer:

A class C amplifier is normally biased beyond cutoff, meaning the transistor is biased so that it conducts for less than half of the input signal cycle. This results in high efficiency but also introduces distortion.

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Question no: 15

What is push pull operation in class B amplifier?

Answer:

When the Q-point is selected at cutoff then collector current flows only during the positive half cycle of input signal. It is called class B power amplifier.

"A second transistor operating in class B can be added to the circuit to conduct when the first transistor is not working or conducting. Such operation is called Push-Pull operation."

It is used when high output power and high efficiency is required.

Long question no.01

$90^0 = 90^0$ at very high frequency.

V. Imp LQ 1

7.4: $\boxed{\text{Low-Frequency Response}}$:-Here we consider two cases:

(i)-Gain of a single stage common emitter RC coupled amplifier in mid frequency range (Gain of mid frequency range):-The most commonly used amplifier is common emitter amplifier. We consider a single stage RC amplifier as shown in fig 7.4(a). Its partially simplified equivalent circuit is shown in fig 7.4 (b). To draw the equivalent circuit, we make following assumptions:

Assumptions:-

(i)-We assume that C_E is large and represents negligibly small reactance compared to R_E i.e $R_E >> X_{CE}$. As C_E ll to R_E, this implies that we can drop R_E and C_E from equivalent circuit.

(ii)-We choose R_1 and R_2 so that $R_B \gg R_{i_2}$ where,

Equivalent Resistance of R_1 & R_2 $R_B = \frac{R_1 R_2}{R_1 + R_2}$

R_{i_2} is input resistance of transistor usually h_{ie}. As $R_B \| R_{i_2}$, so

Here $\boxed{R_B >> R_{i_2}}$ $\frac{R_B R_{i_2}}{R_B + R_{i_2}} \cong R_{i_2}$ Equivalent resistance of R_B and R_{i_2}

Therefore we can drop R_B from equivalent.

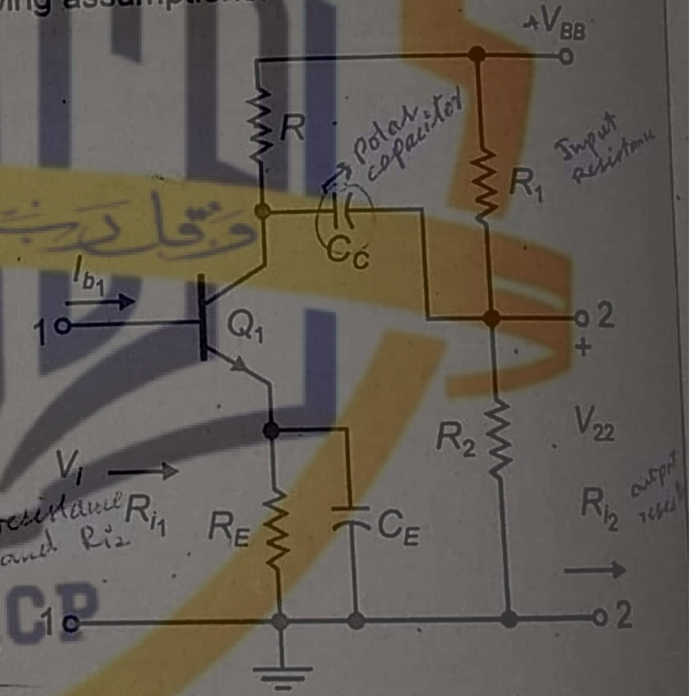

Fig 7.4 (a): One stage of RC amplifier

(iii)-The reactance of coupling capacitor C_C is so small in the mid frequency range that it can be neglected as compared to other reactances of the circuit.

Now applying above assumptions, the equivalent circuit takes the form as shown in fig 7.4 (c).

Writing model equation at point A in fig 7.4 (c), we get

$$\frac{V_{22}}{R_{i_2}} + \frac{V_{22}}{R} + g_m V_{be} = 0$$

$$\Rightarrow V_{22}\left(\frac{1}{R_{i_2}}+\frac{1}{R}\right) = -g_m V_{be}$$

$$\Rightarrow V_{22}\left(\frac{R+R_{i_2}}{RR_{i_2}}\right) = -g_m V_{be}$$

$$\Rightarrow \frac{V_{22}}{V_{be}} = -\frac{g_m R R_{i_2}}{R+R_{i_2}}$$

The mid frequency gain for single stage CE-amplifier is,

$$A_{v(mid)} = \frac{V_{22}}{V_{be}} = -g_m\left(\frac{RR_{i_2}}{R+R_{i_2}}\right)$$

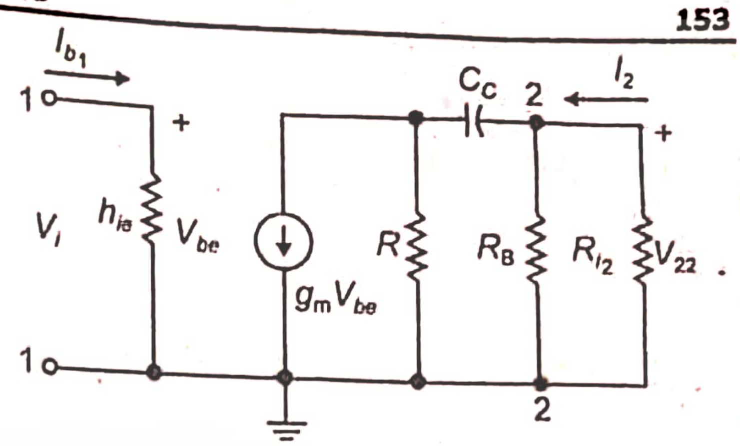

Fig 7.4 (b): Simplified circuit

Negative sign shows that the phase shift between input and output is 180^0. Above equation shows that gain in mid frequency is independent of frequency and is constant in mid frequency.

(i)-Gain of a single stage common emitter RC coupled amplifier in low frequency range (Gain of low frequency range):-The single stage RC amplifier is shown in fig 7.4(a). Its partially simplified equivalent circuit is shown in fig 7.4 (b). To draw the equivalent circuit, we make following assumptions:

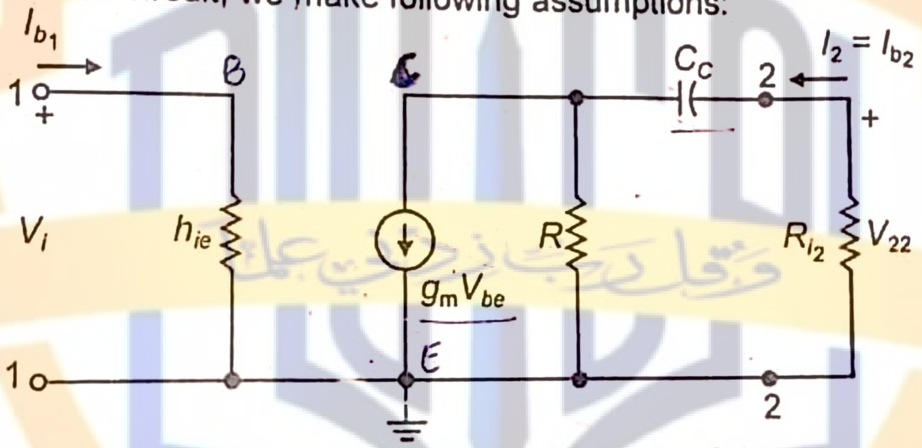

Fig 7.4 (c): g_m model equivalent circuit at low frequency

(i)- We assume that C_E is large ($\cong 50\mu F$) and represents negligibly small reactance in the low frequency region. As C_E ll to R_E, thus we can drop R_E and C_E from equivalent circuit.

(ii)-We choose R_1 and R_2 so that $R_B \gg R_{i_2}$ where, → Here, in low frequency range, we don't neglect Cc.

$$R_B = \frac{R_1R_2}{R_1+R_2}$$

R_{i_2} is input resistance of transistor usually h_{ie}. As $R_B \| R_{i_2}$, so

$$\frac{R_BR_{i_2}}{R_B+R_{i_2}} \cong R_{i_2}$$

Therefore we can drop R_B from equivalent.

(iii)-For low frequency region, coupling capacitor offers appreciable reactance $\left(X_C = \frac{1}{\omega C_C} = \frac{1}{2\pi f C_C}\right)$ and is not negligible as compared to R_{i_2}. So we consider the coupling capacitor in the equivalent circuit.

Hence with above assumptions, the equivalent circuit takes the form as shown in fig 7.4 (c). As C_C and R_{i_2} are in series, therefore, let Z be equivalent impedance of X_{Cc} and R_{i_2}. Then

$\frac{V_{22}}{Z} + \frac{V_{22}}{R} - g_m V_{be} = 0 \Rightarrow \frac{V_{22}}{Z} + \frac{V_{22}}{R} = g_m V_{be} \Rightarrow V_{22}\left(\frac{R+Z}{RZ}\right) = g_m V_{be} \Rightarrow \frac{V_{22}}{Z}$

(due to opposite arrow)

$\left(I_2 = \frac{V_{22}}{Z} = \frac{g_m V_{be} R}{R+Z}\right) \Rightarrow I_2 =$

$$Z = R_{i_2} + X_{C_C} \quad \Rightarrow Z = R_{i_2} + \frac{1}{j\omega C_C}$$

Now the equivalent circuit is shown in fig 7.4 (d). Current through Z is I_2 and is given by,

$$I_2 = g_m V_{be}\left(\frac{R}{R + R_{i_2} - \frac{j}{\omega C_C}}\right) \quad \Rightarrow I_2 = g_m V_{be}\frac{R}{R + R_{i_2}}\left(\frac{1}{1 - \frac{j}{\omega C_C(R + R_{i_2})}}\right)$$

Taking complex conjugate of above equation,

$$I_2^* = g_m V_{be}\frac{R}{R + R_{i_2}}\left(\frac{1}{1 + \frac{j}{\omega C_C(R + R_{i_2})}}\right)$$

Now, $\because j^2 = -1$

$$I_2^2 = I_2 I_2^* = g_m V_{be}\frac{R}{R + R_{i_2}}\left(\frac{1}{1 - \frac{j}{\omega C_C(R + R_{i_2})}}\right) \times g_m V_{be}\frac{R}{R + R_{i_2}}\left(\frac{1}{1 + \frac{j}{\omega C_C(R + R_{i_2})}}\right)$$

$$\Rightarrow I_2 = g_m V_{be}\frac{R}{R + R_{i_2}}\frac{1}{\sqrt{1 + \frac{1}{\{\omega C_C(R + R_{i_2})\}^2}}}$$

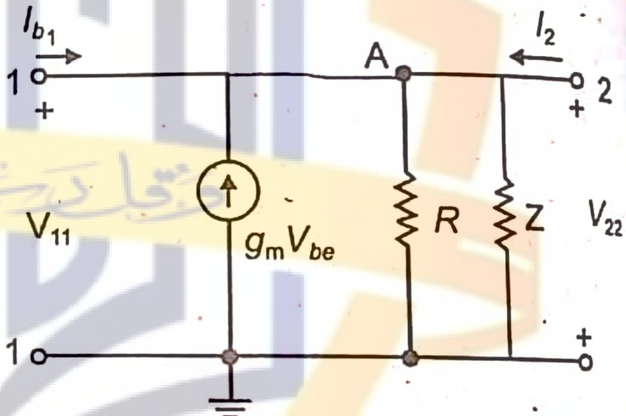

Fig 7.4 (d): The equivalent circuit

The output voltage is,

$$V_{22} = -I_2 R_{i_2}$$

$$= -g_m V_{be}\frac{R R_{i_2}}{R + R_{i_2}}\frac{1}{\sqrt{1 + \frac{1}{\{\omega C_C(R + R_{i_2})\}^2}}}$$

Low frequency gain is,

From previous $\hookrightarrow A_{v(mid)}$

$$A_{v(low)} = \frac{V_{22}}{V_{be}}$$

$$= -g_m\frac{R R_{i_2}}{R + R_{i_2}}\frac{1}{\sqrt{1 + \frac{1}{\{\omega C_C(R + R_{i_2})\}^2}}}$$

The gain ratio is,

$$\frac{A_{v(low)}}{A_{v(mid)}} = \frac{1}{\sqrt{1 + \frac{1}{\{\omega C_C(R + R_{i_2})\}^2}}} \quad \rightarrow (7.1)$$

This equation shows that gain ratio depends upon frequency and $A_{v(low)} < A_{v(mid)}$. When $f = 0$, then $\omega = 2\pi f = 0$ and we have

$$\frac{A_{v(low)}}{A_{v(mid)}} = \frac{1}{\sqrt{1+\infty}} = 0$$

As the frequency increases, the second term in the radical decreases w. r. t. 1and gain ratio increases towards the mid frequency value of unity. There is an associated phase angle in addition to the 180^0 phase shift in $A_{v(mid)}$ given by,

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$$\theta = tan^{-1}\left(\frac{1}{\omega C_C(R+R_{i_2})}\right)$$

At low frequency, the total phase shift will then be $180^0 + \theta_{low}$.

$$\theta_{low} = tan^{-1}\left(\frac{1}{0}\right) = tan^{-1}(\infty) = 90^0$$

At $f = 0$, the total phase shift becomes $180^0 + 90^0 = 270^0$.

7.4.1: The Low Frequency Limit:-Low frequency limit f_1 can be determined by using expression for gain at low frequency range. The gain ratio is,

$$\frac{A_{v(low)}}{A_{v(mid)}} = \frac{1}{\sqrt{1+\frac{1}{\{\omega C_C(R+R_{i_2})\}^2}}} = \frac{1}{\sqrt{1+\frac{1}{\{2\pi f C_C(R+R_{i_2})\}^2}}} \rightarrow (7.2)$$

For low limit frequency $f = f_1$,

$$A_{v(low)} = \frac{A_{v(mid)}}{\sqrt{2}} \quad \Rightarrow \frac{A_{v(low)}}{A_{v(mid)}} = \frac{1}{\sqrt{2}}$$

by comparing

$$\Rightarrow \frac{1}{\sqrt{1+\frac{1}{\{2\pi f_1 C_C(R+R_{i_2})\}^2}}} = \frac{1}{\sqrt{2}} \quad \Rightarrow \sqrt{1+\frac{1}{\{2\pi f_1 C_C(R+R_{i_2})\}^2}} = \sqrt{2}$$

$$\Rightarrow 1+\frac{1}{\{2\pi f_1 C_C(R+R_{i_2})\}^2} = 2 \quad \Rightarrow \frac{1}{\{2\pi f_1 C_C(R+R_{i_2})\}^2} = 2-1$$

$$\Rightarrow \frac{1}{\{2\pi f_1 C_C(R+R_{i_2})\}^2} = 1 \quad \Rightarrow \frac{1}{2\pi f_1 C_C(R+R_{i_2})} = 1 \quad \Rightarrow f_1 = \frac{1}{2\pi C_C(R+R_{i_2})}$$

The lower limit frequency depends upon the size of blocking capacitor and resistance through which it charges. Putting value of f_1 in equation (7.2), we get

$$\frac{A_{v(low)}}{A_{v(mid)}} = \frac{1}{\sqrt{1+\left(\frac{f_1}{f}\right)^2}} \longrightarrow A \quad \text{with phase angle } \theta = tan^{-1}\left(\frac{f_1}{f}\right)$$

At $f = f_1$, phase angle is,

$$\theta = tan^{-1}\left(\frac{f_1}{f_1}\right) = tan^{-1}(1) = 45^0$$

The phase shift increases by 45^0 w. r. t. phase at mid frequency. At $f = f_1$, total phase shift becomes $180^0 + 45^0 = 225^0$.

Bode Plot (Low Frequency Response Curve):-A general curve for low frequency response of all RC amplifiers is plotted with the help of equation A.

$$\left\{\frac{A_{v(low)}}{A_{v(mid)}}\right\}_{dB} = 20log_{10}\left(\frac{1}{\sqrt{1+\left(\frac{f_1}{f\downarrow}\right)^2\uparrow}}\right)\downarrow$$

This equation shows that when ratio (f_1/f) goes on increasing (*i.e.* when frequency f goes on decreasing), the gain $A_{v(low)}$ goes on decreasing. To plot gain versus frequency, we find the gain ratio for following different values of (f_1/f).

$$\frac{f_1}{f} = 1,10,100,1000$$

These values have been so chosen that there is a change of 10 times from one ratio to the other. This is known as change of one decade.

1 = For $\frac{f}{f_1} = 1$ i.e. $\frac{f_1}{f} = 1$, $\left\{\frac{A_{v(low)}}{A_{v(mid)}}\right\}_{dB} = 20log_{10}\left(\frac{1}{\sqrt{1+1}}\right) = -3dB$

0.1 = For $\frac{f}{f_1} = \frac{1}{10}$ i.e. $\frac{f_1}{f} = 10$, $\left\{\frac{A_{v(low)}}{A_{v(mid)}}\right\}_{dB} = 20log_{10}\left(\frac{1}{\sqrt{1+(10)^2}}\right) = -20dB$

0.01 = For $\frac{f}{f_1} = \frac{1}{100}$ i.e. $\frac{f_1}{f} = 100$, $\left\{\frac{A_{v(low)}}{A_{v(mid)}}\right\}_{dB} = 20log_{10}\left(\frac{1}{\sqrt{1+(100)^2}}\right) = -40dB$

0.001 = For $\frac{f}{f_1} = \frac{1}{1000}$ i.e. $\frac{f_1}{f} = 1000$, $\left\{\frac{A_{v(low)}}{A_{v(mid)}}\right\}_{dB} = 20log_{10}\left(\frac{1}{\sqrt{1+(1000)^2}}\right) = -60dB$

A general curve for low frequency response of all RC amplifiers is plotted in fig 7.5. In order to show the gain increasing with frequency as a normal low frequency response, the curve is plotted in terms of f/f_1.

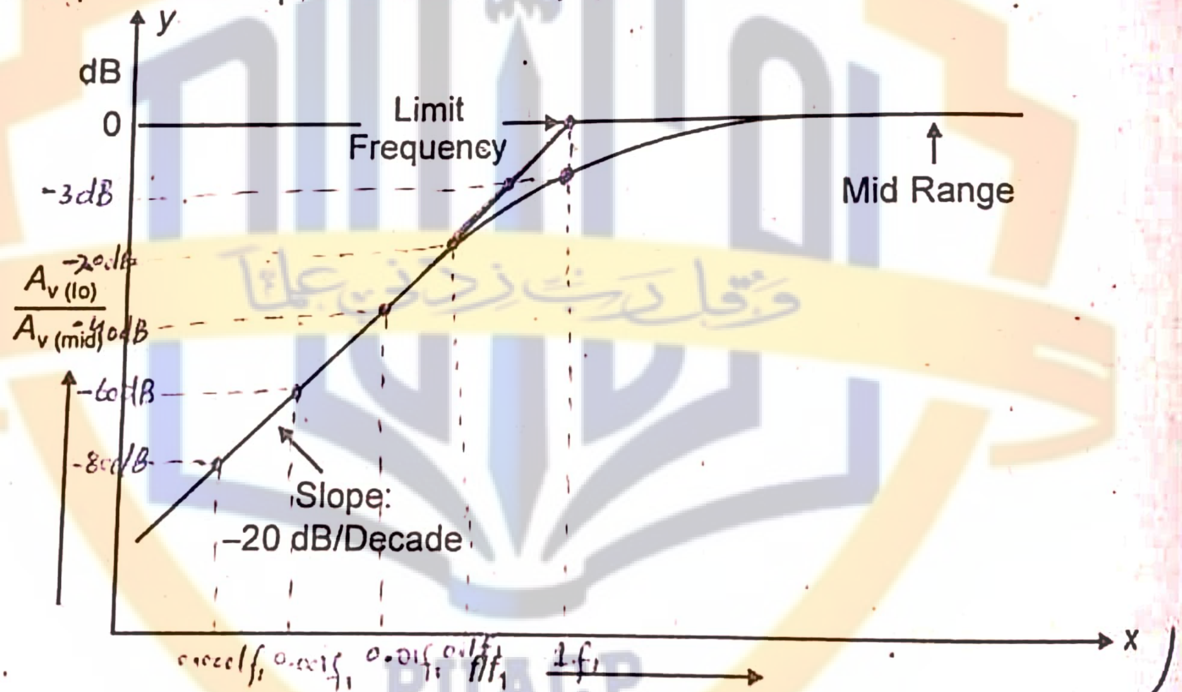

Fig 7.5: Low frequency response in decibels

7.5: The Unbypassed Emitter Resistor (Gain of Common Emitter Amplifier with Unbypassed Emitter Resistor):

-Circuit of common emitter amplifier with bypass emitter resistor is shown in fig 7.6 (a). Its equivalent circuit is shown in fig 7.6 (b). Voltage equation for input loop is,

$$V_i = h_{ie}I_b + (I_b + I_c)R_E$$

Putting $I_c = h_{fe}I_b$, we get

$$V_i = h_{ie}I_b + (I_b + h_{fe}I_b)R_E \quad \Rightarrow V_i = I_b\{h_{ie} + (1+h_{fe})R_E\} \longrightarrow (1)$$

h_{ie} is the input resistance of transistor and I_b is input current.

Input Resistance:-The input resistance of circuit is given by,

$$R_i = \frac{V_i}{I_b} = \frac{\cancel{I_b}\{h_{ie} + (1+h_{fe})R_E\}}{\cancel{I_b}} \quad \Rightarrow R_i = h_{ie} + (1+h_{fe})R_E$$

$$\Rightarrow R_i = h_{ie} + h_{fe}R_E \qquad \because h_{fe} \gg 1$$

This is expression for input resistance.

Now

$$I_b = \frac{V_i}{h_{ie} + (1 + h_{fe})R_E} \cong \frac{V_i}{h_{ie} + h_{fe}R_E} \qquad \because h_{fe} \gg 1$$

Also

$$V_{be} = h_{ie}I_b = \frac{h_{ie}V_i}{h_{ie} + h_{fe}R_E} = \frac{V_i}{1 + \frac{h_{fe}}{h_{ie}}R_E} = \frac{V_i}{1 + g_mR_E} \qquad \Rightarrow V_i = (1 + g_mR_E)V_{be}$$

This equation gives input voltage in the presence of R_E. In the absence of R_E, input voltage is,

$$V_i = V_{be}$$

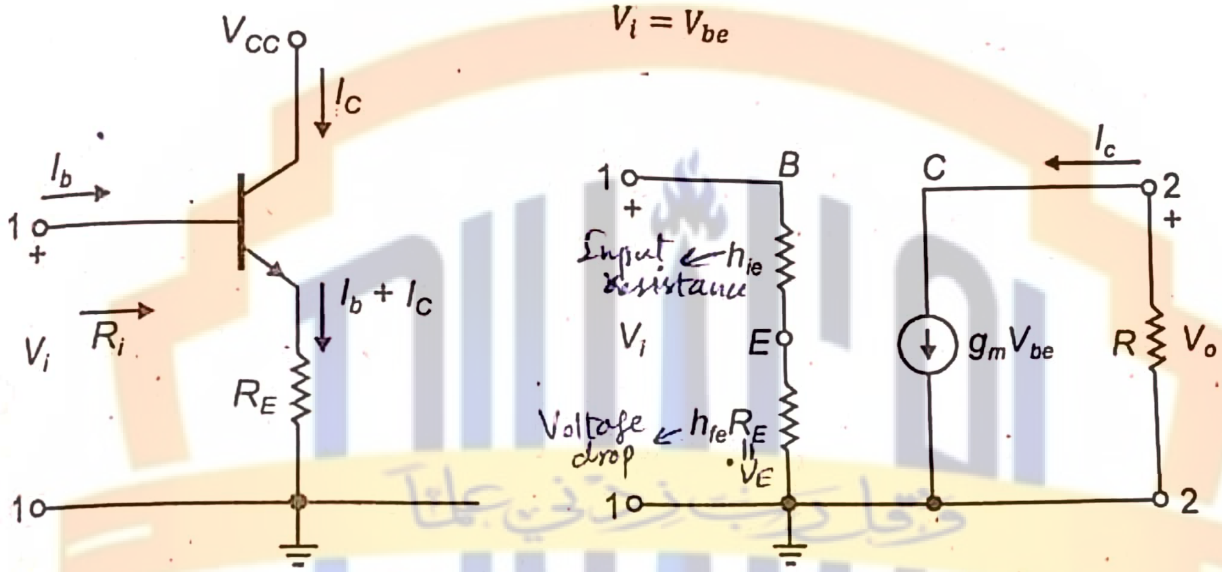

Fig 7.6 (a): The Unbypassed Emitter Resistor Fig 7.6 (b): Equivalent circuit

Output Voltage:-From equivalent circuit, the output voltage is,

$$V_0 = -I_cR = -h_{fe}I_bR = -h_{fe}\left(\frac{V_{be}}{h_{ie}}\right)R = -\left(\frac{h_{fe}}{h_{ie}}\right)V_{be}R$$

$$\Rightarrow V_0 = -g_mRV_{be} = -g_mR\left(\frac{V_i}{1 + g_mR_E}\right) = A_{v(mid)}\left(\frac{V_i}{1 + g_mR_E}\right)$$

Voltage Gain:-Voltage gain for common emitter amplifier without R_E (i.e. $R_E = 0$) is given by relation,

$$A_v = \frac{V_0}{V_i} = -\frac{g_mRV_{be}}{V_{be}} = -g_mR$$

Voltage gain for common emitter amplifier with R_E is given by relation,

$$A'_v = \frac{V_0}{V_i} = -\frac{g_mRV_{be}}{(1 + g_mR_E)V_{be}} = \frac{(-g_mR)}{1 + g_mR_E} = \frac{A_v}{1 + g_mR_E}$$

This equation shows that the gain is reduced by the presence of R_E. V.V. Same results can be obtained for the gain with an unbypassed source resistance R_E for an FET or for an unbypassed cathode resistance R_k for a tube.

Imp 7.6: High-Frequency Equivalent Circuits and The Miller Effect:-At high frequency, the transistor has two pn-junction capacitances:

(i)- C_{be} (base to emitter capacitance):-Diffusion region capacitance when the junction is forward bias. It holds the charge stored in the base and is of many pico farads.

QNO (02)

Briefly discuss the way to connect negative feedback with operational amplifiers. What are the benefits of negative feedback in operational amplifiers?

Answer:

(From Notes)

Benefits:

The benefits of negative feedback in operational amplifiers are following.

It improves the frequency response.
It reduces the distortion and noise in the amplifier.
It stabilizes the gain of an amplifier.
It increases the bandwidth of the amplifier
It makes the circuit or system self-correcting

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QNO(03)

What is oscillator? Discuss working principle of oscillator. Name the two types of oscillator.

Answer:

Oscillator:

"The circuit which converts DC energy into AC energy at a very high frequency is known as an oscillator."

connected. The frequency of output signal is determined by passive component used in the oscillator and can be varied at will.

Types of oscillators: Electronic oscillators may be divided into following two classes:
(a) Harmonic
(i)-Sinusoidal oscillators – which produce an output having sine waveform.
(b) Relaxation
(ii)-Non-sinusoidal oscillators – they produce output which has square, rectangular or saw-tooth waveform or of pulse shape.

Sinusoidal oscillators may be further subdivided into following types:
(1)-Crystal oscillators such as Pierce oscillator.
(2)-Negative resistance oscillators such as tunnel diode oscillator.
(3)-Tuned circuits or LC feedback oscillators such as Hartley, Colpits and Clapp etc.
(4)-RC phase shift oscillators such as Wien-bridge oscillator.
(5)-Heterodyne or beat-frequency oscillator (BFO).

Imp **10.2: Oscillator Feedback Principles**:-From the voltage feedback circuit, we have

$$\beta = \frac{V_f}{V_0} = \frac{V_i}{V_0} \qquad \Rightarrow V_f = V_i = \beta V_0$$

Since $V_0 = -AV_i$, so

$$V_i = -\beta A V_i \qquad \Rightarrow V_i + \beta A V_i = 0$$

$$\Rightarrow V_i(1 + \beta A) = 0$$

If output is present, then $V_i \neq 0$ and we have

$$1 + \beta A = 0 \qquad \Rightarrow \beta A = -1$$

S/Q This is the condition of oscillation in a feedback amplifier.

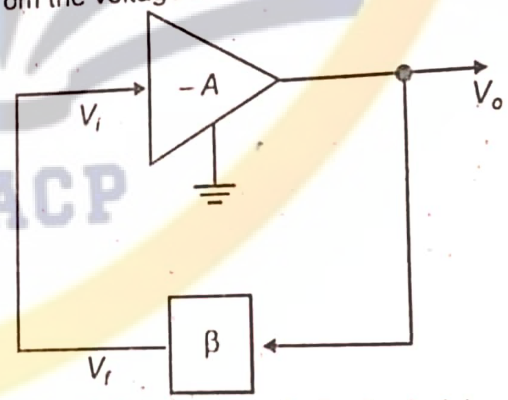

Fig 10.3: Oscillator feedback principles

Hence two requirements for oscillations to occur are:
(i)- $\beta A = -1$
(ii)-The net phase shift around the feedback loop be 0^0 or 360^0 or $2n \times 180^0$, where $n = 0,1,2,3,\cdots$.

The amplifier provides its own input and initially the gain must be such that $|\beta A| > 1$. An initial turn on surge or noise voltage provides V_i at the input and this is amplified to the output. This amplified out put is fed back to the input as a larger signal. The process is repeated at successively greater amplitudes until the output becomes limited at cut off and saturation. By operation to those limits the gain is reduced to an average level and a steady level is maintained. The frequency of oscillation adjusts itself so that the phase shift requirement is satisfied.

10.3: The Hartley Oscillator:-The main

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Oscillator feedback principle

The circuit for the oscillator fed back output voltage to input voltage as shown in the figure,

From the figure, we have

$$\beta = \frac{V_f}{V_o'}$$

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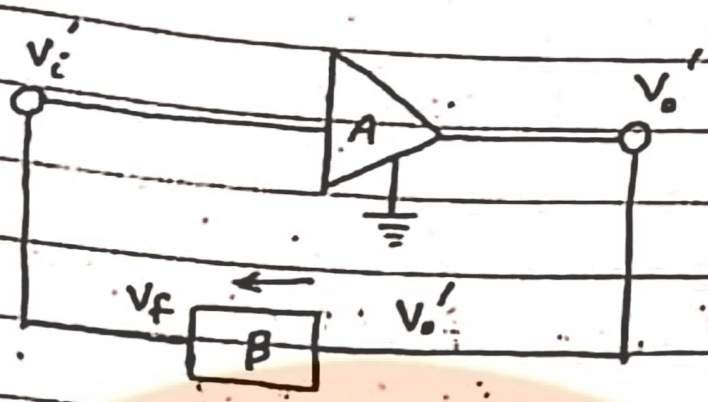


$$V_f = \beta V_o'$$

But $V_f = V_i'$

$$V_i' = \beta V_o' \Rightarrow (1)$$

and voltage gain of Amplifier is $A = -\frac{V_o'}{V_i'}$

The negative sign for one 180° phase shift between the output and input.

$$V_o' = -AV_i'$$

using in (1)

$$V_i' = -A\beta V_i'$$

$$\therefore V_i' + A\beta V_i' = 0$$

$$V_i'(1 + A\beta) = 0$$

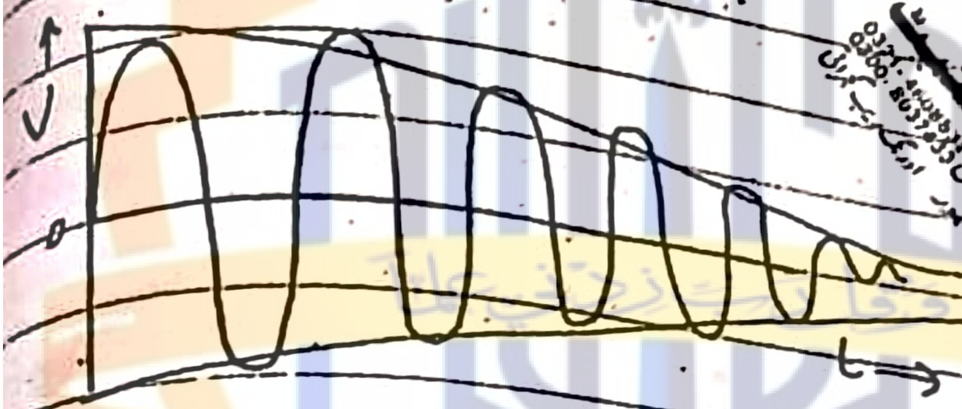
if $V_i = 0$ then there is no amplification, so,

$$1 + A\beta = 0$$

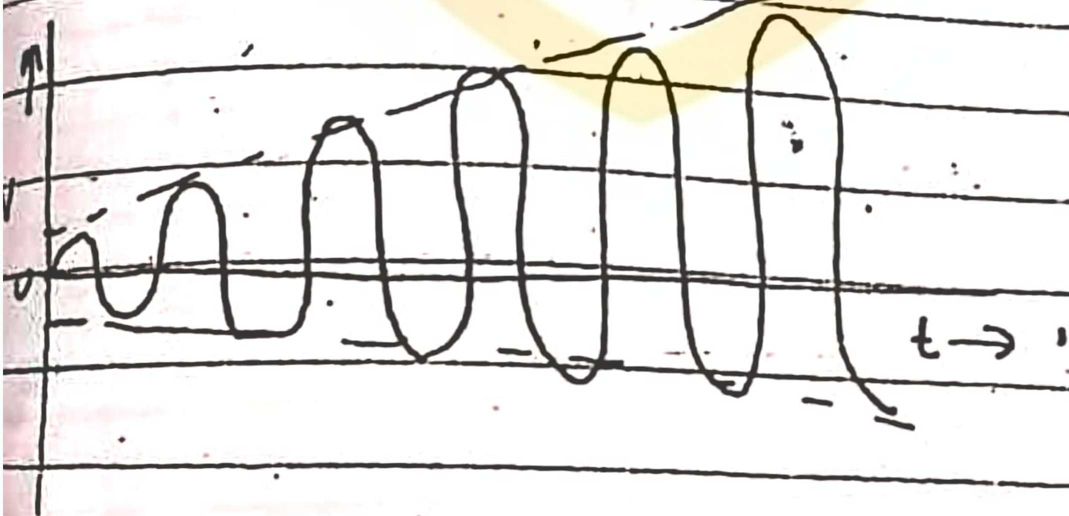
$$\Rightarrow A\beta = -1$$

Now, we have the following conditions on the $|A\beta|$.

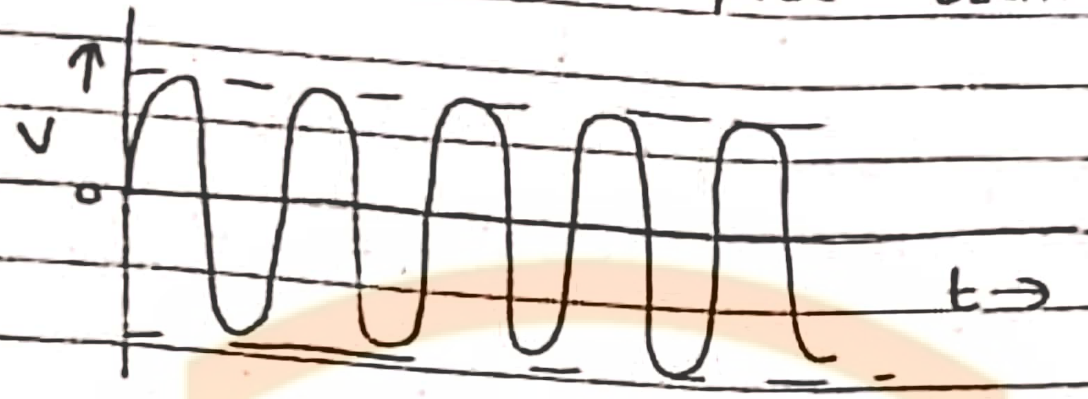
1) when $|A\beta| < 1$ then we have damped oscillation as shown in figure.



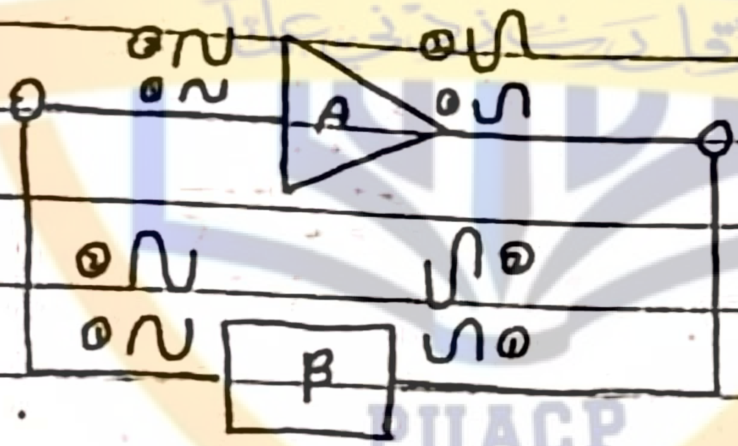
2) If $|A\beta| > 1$ then output signal is build up is known as growing oscillator.



③ If $A\beta = -1$ then we has sustained undaped oscillations



Let us consider a fet back oscillator as shown in figure.

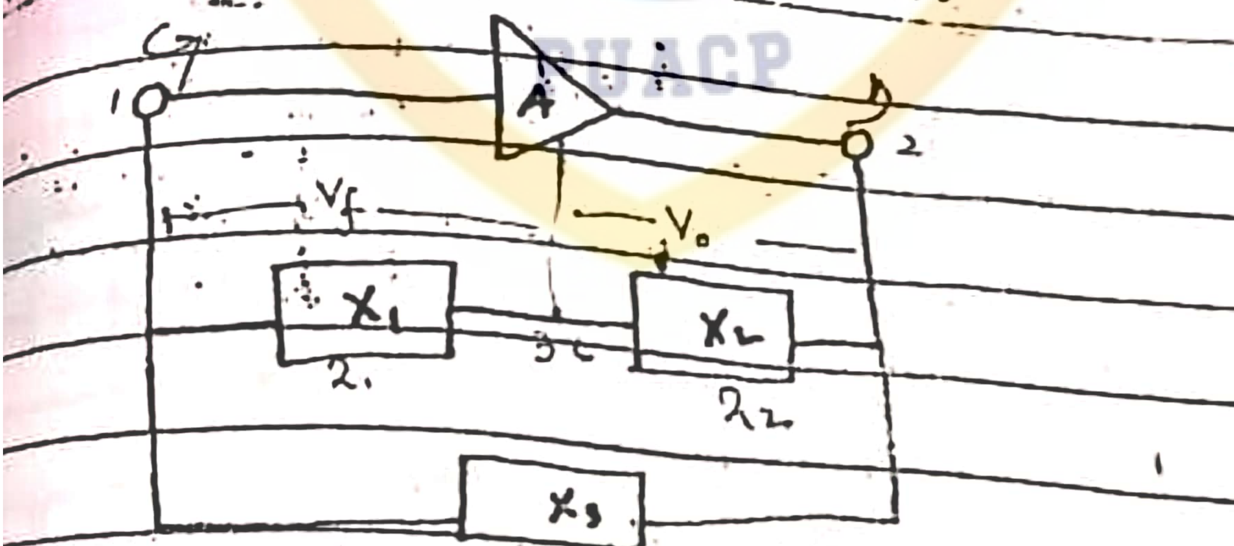


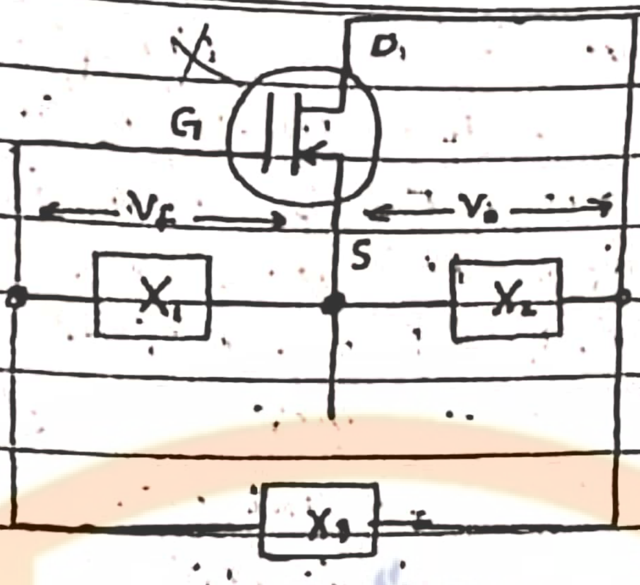
This take the small portion of output and through β-network it take to the input of the amplifier again.

This process is continuous and we have un level oscillation at the output because it Ap amplifie it more and more. so, to get the desirable oscillation we must do amendment in place of β- network.

Amplitude limitation by cut off and saturation implies distortion and the resultant Harmonic frequency is filter by the resonanto circuit in place of β- network.

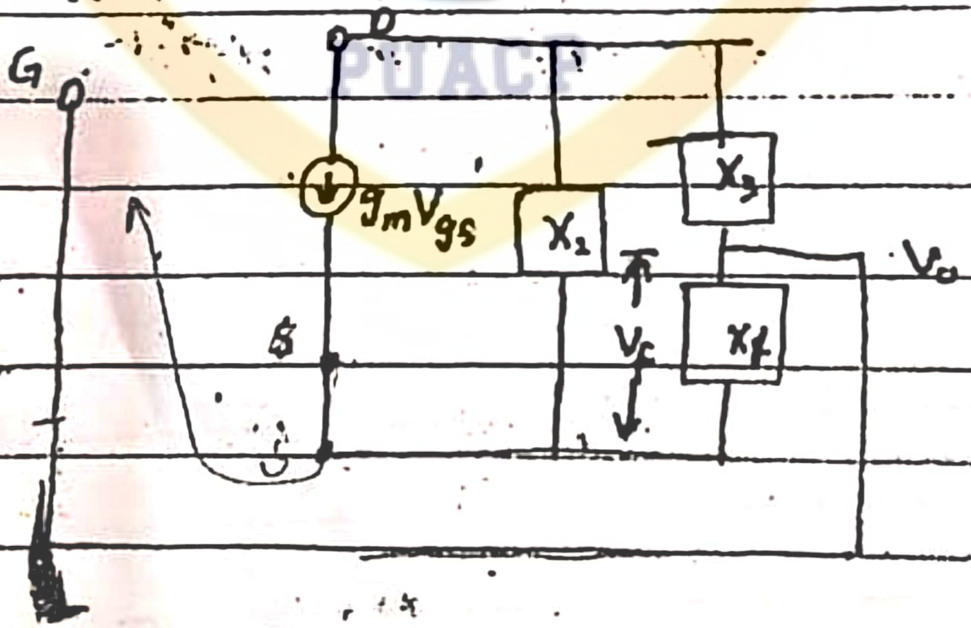
Thus, the circuit for better undistortion Amplifier is shown in the figure.





The reactive circuit provide the 180° phase shift of the input and output voltage Also, the FET has 180° additional phase shift; which required the second condition for oscillation.

Now, the equivalent circuit for the above is shown.



Types of oscillator:

Two main types of oscillator are LC and RC oscillator.

LC Oscillator:

It is the oscillatory circuit which consists of two elements i.e; an inductor and a capacitor. Both are capable of storing energy. The capacitor stores energy in the form of electric field when there is potential difference between its plates. Similarly a coil or inductor stores energy in its magnetic field whenever current flows through it. Both L and C supposed to be loss free.

⇒ LC oscillator is further classified into following types.

1) Hartley oscillator
2) Colpitts oscillator
(3) Tuned collector oscillator

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RC oscillator:

It is a type of oscillator that uses resistors and capacitors in a feedback network to generate an oscillating signal.

→ It typically consists of an amplifier and a RC network that provides the necessary phase shift for oscillation to occur. This type of oscillator is commonly used to produce sinusoidal signal at low to medium frequencies.

⇒ RC oscillator is further classified into following types.

1) **Phase-shift oscillator**

2) **Wien-Bridge oscillator**

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